

Investigating Avatar Facial Expressions and Collaboration Dynamics for Social Presence in Avatar-Mediated XR Remote Communication

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ABSTRACT

In avatar-mediated remote communication, reflecting one's facial expressions in avatars remains challenging but essential in facilitating effective communication and fostering social presence. From our prior studies, we underscored the significance of emotion-based facial expressions in avatar-mediated remote communications within eXtended Reality (XR). Our findings indicated that during emotional conversations as well as informative speeches, activating the emotion-based blendshapes resulted in a level of social presence and communication quality comparable to activating all facial blendshapes. As we move forward, our research aims to investigate avatar facial expression considering XR contexts, with an emphasis on the challenges stemming from XR device constraints and the dynamic contexts of remote collaboration. We aspire to devise adaptive guidelines for avatar facial expression, leveraging emotional cues using XR devices. Additionally, we plan to systematically analyze the implications of remote collaboration's varying scenarios, formulating guidelines for adaptive avatar representation that account for different collaborative environments and scenarios. To enhance social presence and communication experiences among users, we are preparing for a series of user studies, aiming to validate the effectiveness of our approaches and prototype systems.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Collaborative and social computing—Empirical studies in collaborative and social computing

1 RESEARCH BACKGROUND

In this age of pervasive eXtended Reality (XR) technologies, remote communication has taken on a new dimension with the use of embodied virtual avatars featuring realistic human appearances [4]. In avatar-mediated communication, facial expressions are essential elements to express and understand a person's emotional state, facilitating effective communication [3]. Acknowledging the importance of avatar facial expressions, various approaches have been developed to improve the expressiveness of avatar's face. Though previous works have attempted to tackle real-time facial processing challenges through different approaches, they often overlook the emotional and communicative perspectives that are particularly important in communication contexts. In addition, accurately mirroring the intricate details of human facial expressions in avatars remains challenging given the complexity and subtlety of human facial expressions. This has led to a potential gap in the behavioral realism in terms of avatar facial expressions that can affect communication quality [1].

2 RESEARCH PROGRESS

To bridge this research gap, our first research aimed to explore the intricate role of facial combinations and their perceptions in

avatar-mediated communication, particularly when these combinations are simplified or minimized. In the recent paper, we investigated the influence of different facial blendshape combinations on social presence and communication quality in avatar-mediated remote communication scenarios: informative speech and emotional conversation [2]. Our approach includes a systematic analysis of facial blendshapes, leading to the two primary facial combinations: visually noticeable blendshapes, typically observed in casual conversations, and emotion-based, derived from Paul Ekman's FACS. We investigated the effects of these facial combinations against a baseline of basic expressions and the full set of available blendshapes (see Figure 1). The conducted user study ($n = 32$) underscores the crucial role of emotion-based blendshapes, suggesting that the emotionally essential blendshape combination outlines levels of social presence and communication quality. Our research highlights the potential of a streamlined approach for avatar facial control, facilitating emotional expressions and optimizing resources, fostering an avatar-mediated communication experience.

3 RESEARCH OBJECTIVE

Building upon our prior work that underscored the importance of emotion-based facial expressions in avatar-mediated remote communication, we have remaining challenges in mind. Our future research is set to delve deeper into representing avatar facial expressions within the context of XR devices and the dynamics of remote collaboration. As we understand that device constraints and the need for adaptability in diverse remote collaboration scenarios present unique challenges, we aim to develop adaptive guidelines for avatar facial expression in avatar-mediated remote collaborations.

Furthermore, we plan to carry out a systematic analysis that takes into account the varied contexts and conditions of remote collaboration. Through this, we aim to craft a set of avatar design guidelines that are adaptive to the requirements of different collaborative environments and interaction contexts. By doing so, we seek to enhance the emotional expression and communicative effectiveness of avatars, ensuring that non-verbal cues are well represented even in the constraints of current XR technologies.

With the constraints and challenges in mind, we structure our research around the following research questions:

- RQ1.** How can we enhance avatar facial expressions to convey a broad range of human emotions effectively using XR devices in avatar-mediated remote collaboration?
- RQ2.** How do the effects of different facial combination differ by varying user environments and scenarios?
- RQ3.** What are the adaptive avatar representation guidelines to accommodate the different collaborative contexts in XR?

4 RESEARCH PLAN

To address the research questions outlined, our research plan is structured into two distinct phases. In the initial phase, we are focusing on the systematic analysis of avatar facial expressions across different remote collaboration scenarios. Especially, the need for research arises from the creation of shared spaces for remote collaboration, where the size and type of shared objects (e.g., tables) can influence

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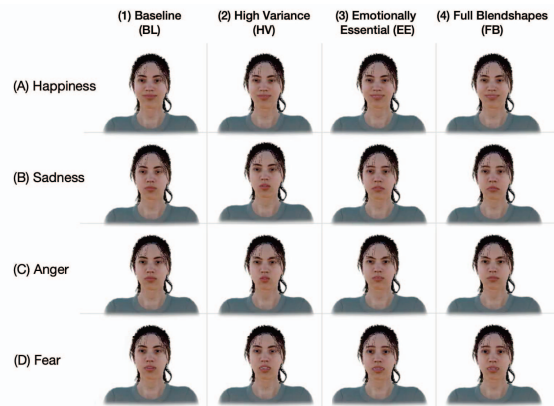


Figure 1: Facial expressions used in the experiment: (A) Happiness, (B) Sadness, (C) Anger, (D) Fear; (1) Baseline (BL), (2) High Variance (HV), (3) Emotionally Essential (EE), and (4) Full Blendshapes (FB).

the perceived distance between collaborators. As previous studies show that the perceived distance can affect emotion recognition, we will investigate how distance, varied nonverbal cue (e.g., gestures), and the extent of different facial expressions impact social presence and communication quality in remote collaboration scenarios. We aim to fill the research gap by studying the combined effects of spatial dynamics, nonverbal communication, and facial expressiveness on the overall effectiveness of remote collaborative interactions. The ultimate aim of this phase is to suggest the guidelines into an adaptive framework that can be readily adapted for various XR applications, enhancing the overall quality of avatar-mediated communication.

The second phase aims to develop adaptive representation guidelines for avatar facial expressions considering the constraints of XR devices. This will involve a detailed investigation into the emotion analysis for supplementing limitations imposed by various XR technologies on avatar expressiveness. Recognizing that certain facial regions may be obscured during XR interactions, we will explore alternative emotional cues that can compensate for these obstructions. We will prototype and iterate on facial expression guidelines that maintain the integrity and effectiveness of emotional communication. This phase will result in a set of preliminary guidelines that are grounded in empirical research and user feedback, ensuring they are robust enough for practical application in diverse XR settings.

5 DISCUSSION POINTS

Given our current research progress and plans, we seek the opportunity to discuss and receive advice on the following points during this doctoral consortium.

One of the core discussion points focuses on representing avatar facial expressions within the limitations of current XR technology. As we mentioned earlier, capturing the minute details of human facial expressions to accurately represent them in virtual avatars is challenging given the complexity and subtlety of human facial expressions. Additionally, all the users participating in the collaboration have their own different XR devices and computational contexts. Therefore, the adaptability of avatars to compensate for hardware and computational constraints without compromising the quality of social interaction is important.

To convey a broad range of emotions effectively in avatar-mediated remote communication, the challenge lies in identifying which facial cues are essential for emotional expressiveness. Thus, we need some discussions on enhancing avatars' facial expressions through comprehensive emotion analysis from limited data, and investigating how people will perceive the partner avatar's facial expression when the capabilities of XR devices are limited.

Another discussion point relates to the user's perception and the efficacy of communication through avatars that are animated with

full versus emotion-based facial blendshapes. By conducting comparative studies across a spectrum of devices and communication scenarios, we aim to investigate how different levels of facial expression affect the user's sense of social presence. These investigations will inform the development of practices for avatar design in XR, particularly in terms of generating emotionally expressive avatars that are optimized for social interaction.

Furthermore, the research will discuss the formulation of adaptive avatar representation guidelines for various remote collaboration contexts. This involves a multi-faceted approach that considers the technical aspects of avatar creation, such as the fidelity of expressions, as well as the dynamic social and task-oriented scenarios of virtual interaction. The discussion will extend to the application of these guidelines in different scenarios, such as formal business meetings versus casual social gatherings, and how avatar expressiveness can be achieved to enhance communication effectiveness in these distinct XR settings. By integrating these findings, the research aims to establish a comprehensive framework for avatar expressiveness that is both adaptive and user-centric.

6 CONCLUSION

In this paper, we outlined some discussion points and areas for improvement to be brought forth in the Doctoral Consortium. In summary, compensating for hardware limitations to provide a rich emotional experience, we are exploring how avatar facial expressions can be made more efficient and accurate in representing a wider array of emotional expressions. To refine our adaptive avatar representation guidelines, we focus on the development of a dynamic facial expression framework that can seamlessly adjust to varying communication contexts and user preferences. We aspire to create guidelines that are not only technically sound but also emotionally engaging, accommodating the dynamics of human interaction. Furthermore, we recognize the importance of interdisciplinary insights – from psychology to computer science – to enrich these guidelines.

In the consortium, we aim to engage with the aforementioned questions and improvement areas. We seek to foster discussions that will not only examine our current methodology and findings but also provide constructive pathways that could lead to breakthroughs in avatar-mediated communication. Through collaborative inquiry and expert feedback, we anticipate refining our approach and identifying new opportunities for innovation in the realm of XR and beyond.

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